

Multiple Choice (10 marks)

1. Which of the following is not a way to form recombinant DNA?
 - (a) Translation
 - (b) Conjugation
 - (c) Specialized transduction
 - (d) Transformation
 - (e) N/A
2. In most ecosystems, net primary productivity is important because it represents the
 - (a) biomass of all producers
 - (b) biomass of all consumers
 - (c) energy available to heterotrophs
 - (d) energy consumed by decomposers
 - (e) total solar energy converted to chemical energy by producers
3. An organism exposed to wild temperature fluctuations shows very little, if any, change in its metabolic rate. This organism is most probably a
 - (a) thermophyle.
 - (b) poikilotherm.
 - (c) ascospore.
 - (d) endotherm.
 - (e) stenotherm.
4. A deficiency in which endocrine gland would result in tetany?
 - (a) Pancreas
 - (b) Thymus
 - (c) pituitary gland
 - (d) adrenal gland
 - (e) thyroid gland
5. What is the probability of getting exactly three heads infive flips of a balanced coin?
 - (a) 12.5 percent
 - (b) 40 percent
 - (c) 18.75 percent
 - (d) 25 percent
 - (e) 37.5 percent

True / False (10 marks)

Answer True or False

6. True or False: Can infundibular height predict the clearance of lower pole calyceal stone after extracorporeal shockwave lithotripsy?
7. True or False: Does blood pressure change in treated hypertensive patients depending on whether it is measured by a physician or a nurse?
8. True or False: Can homemade alcohol (Raksi) be useful for preserving dead bodies?
9. True or False: Can surgeon familiarization with current evidence lead to a change in practice?
10. True or False: Is the covering of the resection margin after distal pancreatectomy advantageous?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the chromosomal composition in an animal with a haploid number of 10, detailing the chromosome and chromatid counts in the spermatogonium, first polar body, second polar body, and secondary oocyte.
12. Discuss the role of positive feedback mechanisms in the endocrine system, using the example of infant suckling and oxytocin release to illustrate your analysis.

End of Paper